sanas South African National Accreditation System	**TECHNICAL AND PRODUCTION REQUIREMENTS of ISO 17034:2016**	Organisation's SANAS No/s.

Date/s of evaluation	
Assessor/s & Observers	
Laboratory	
Area / field of operation	
Laboratory Representative	

This report covers the following:

Document Review only		Implementation on Site Visit only		Document Review and Site Visit		Other	

REQUIREMENTS & COMMENTS. ***Compliance = C, Non-compliance = NC***

NB1: *References to ISO/IEC 17034:2016 are in italics. The order of assessment need not follow the order of the checklist. Assessors are expected to know & have the standard, this worksheet is designed as guidance to prompt detailed recording of the process.*

REFER TO ISO 17034:2016 FOR DETAIL AND FOR CLARIFICATION NOTES.

NOTE 1: *For **CAB's comments**: The CAB must provide information on how requirements have been addressed, documented and/or implemented. Make reference to policies / procedures, incl. clause numbers.*

NOTE 2: *For **Assessor's Comments**: The Assessor must provide information on the CAB's conformity with the requirements*

©SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7. TECHNICAL AND PRODUCTION REQUIREMENTS				
7.1	**General requirements**			
	A CRM has at least one certified value Clause 7.9 applies only to certified values			
7.2	**Production planning:**			
7.2.1	• Has the RMP identified those process that directly affect the quality of the RM?	Production plan ASMF7.2.1 was developed and implemented		
	• Has the RMP planned those process that directly affect the quality of the RM?	Production plan ASMF7.2.1 was developed and implemented		
	• Has the production plan been documented?	Production plan ASMF7.2.1 was developed and implemented		
7.2.2	• Has any technical input from sub-contractors been specified?	No sub-contracting has taken place		
	• Is this technical input documented, and regularly reviewed?	Document ASMF4.1.2 (Checklist) was implemented		
7.2.3	Has the RMP addressed, during the planning sage:-			
a)	Material selection? (including sampling, where applicable)	Is defined in section 1 of the production plan ASMF7.2.1		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
b)	Verification of the identity of the material?			
c)	Maintaining a suitable environment for production?	It is defined in procedure ASMF7.2.3, section 6: Check and record temperature and humidity before starting with work and closely monitor these throughout the shift.		
d)	Material processing?	Is defined in section 3 of Production plan ASMF7.2.1		
e)	Choice of measurement procedures?	Are defined in section 5 of Production plan ASMF7.2.1		
f)	Validation of the measurement procedures?	See the attached method validation on SANAS website of Inherent moisture, CV, Ash%, VM		
g)	Calibration/verification of measuring equipment?	That is defined in respective testing method for coal ie. CV test method, Ash test method, VM test method, IM test method and sample preparation method.		
h)	Specification of acceptance criteria? (homogeneity including sampling)	Is defined in section 3 and 4 of Production plan ASMF7.2.1		
i)	Specification of acceptance criteria? (stability including sampling)	Is defined in section 3 and 4 of Production plan ASMF7.2.1		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
j)	Design and organization of characterization? (sampling)	It is defined in section 4 ASMF 7.12 Characterisation.		
k)	Commutability? (where appropriate)	It is defined in section 6 ASMF 7.15 Distribution and customer support procedure Courier service providers must be chosen in the sense that they are able to provide tracking capabilities and a shorter time frame (defined as < or = 4 days) to deliver to clients.		
l)	Assigning property values?	The detail procedure steps defined in section 5 of procedure ASMF 7.13		
m)	Establishing uncertainty budgets incl. uncertainty of certified values?	It is defined in document ASMF UB001		
n)	Defined acceptance criteria for measurement levels and their uncertainties?	It is defined in document ASMF UB001		
o)	Metrological traceability of measurement results and certified values?	It is defined in ASMF 7.9		
p)	Issuing of RM documents?	Issuing of Reference Material documents(certificate) is as per client purchase in that after a client makes a purchase, a soft copy of the reference material certificate is emailed to the client, and a hard copy accompanies the purchased CRM. This is to ensure that the client receives a correct copy of the certificate and that the material sent to them has no confusion whatsoever that may arise.		

©SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
q)	Ensuring storage facilities and conditions?	It is defined in procedure ASMF7.2.3, section 5 which states the parameter of control, the range to apply the control measure and the respective areas of the laboratory.		
r)	Appropriate labelling and packaging?	It is defined in procedure ASMF7.15		
s)	Transport arrangements?	It is defined in procedure ASMF7.15		
t)	Post production stability monitoring?	The CAB addresses stability monitoring of certified reference materials post production in 7.4 Material Handling and Storage under section 6 post production stability monitoring. This section outlines the study, how it is conducted and future amendments (where applicable) as a reaction to the outcome of the study. See attached Stability Monitoring document as evidence		
u)	Adequacy of post distribution service for RM users?	The CAB has in place 7.15 Distribution and Customer Support Procedure to tend to how material from the lab is to be packed and distributed in order to safeguard the quality and integrity of PT samples and certified reference materials. Where issues arise from the distributions of such materials produced in the lab, clients log a complaint and the lab has 7.18 Complaints and handling procedure to combat any issues that may arise from any of the services the CAB offers. The complaints and appeals procedure is displayed publicly in the CABs website for clients to access when issues arise.		
7.2.4	Where multiple batches of RM with equivalent properties are produced, has the RMP verified that information from previous batches is still applicable to the new batch?	The CAB has outlined in AS001IW PROCEDURE ON SAMPLE PREPARATION OF HARD COAL Rev 02 under 4.1 Rationale , how the assignment of batches to originally produced certified reference materials will be conducted when the need arises.		
7.3	**Production Control**			

©SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	• Has the RMP verified that the production plan has been implemented as specified?	The checklist used during sample preparation, testing and LIMS data is able to provide auditable data for assessment per CRM produced		
	• Have deviations from the plan been documented and approved?	To address deviations , ASMF 8.7.3 improvement document was implemented		
7.4	**Material handling and storage**			
7.4.1	• Has the RMP made arrangements to ensure the integrity of candidate RM and RM's throughout the process?	Material handling and storage ASMF 7.4.1 designed and implemented to maintain stability, quality, identity and traceability through out the handling processes.		
	• Have precautions been taken against adverse environmental conditions during processing?	Yes there's condition monitoring measures applied during testing and storage. Procedure ASMF 7.2.3 is implemented and the recording of condition on AS018TF checklist and AS023TF checklist		
7.4.2	Has the RMP identified, preserved and separated candidate RM and RM's from chemicals and other samples, throughout the process?	The packaging of CRM's are done with plastic pulped containers that have a seal proof lid.		
7.4.3	• Has the RMP used adequate packaging?	Yes and we are continually applying stability checks to measure deterioration of the quality of CRM		
	• Have following been considered for the packaging of the RM's: light-shielding, air-free, moisture-free or inert gas packaging?	Yes these conditions are mentioned ASMF 7.15 Distribution and customer support procedure section packaging and distribution. Tamper proof packaging is a requirement under this section.		
	• Has the RMP secure areas or stock rooms which will prevent damage or deterioration?	Yes**:** condition monitoring measures applied during testing and storage. Procedure ASMF 7.2.3 is implemented and the recording of condition on AS018TF checklist and AS023TF checklist		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.4.4	Is the condition of RM's assessed at appropriate intervals to detect possible deterioration?	• After production of a Reference material, the material is assigned a shelf life of 1 year. During the course of the year, on the 25th of each month the CAB conducts a stability monitoring of certified material. The acceptability criteria is set as follows: The average of each month's monitoring must be reproducible to the original assigned value. • See attached stability monitoring document as proof of work.		
7.4.5	• Are the packaging and labelling processes controlled, and do they conform to safety and transport requirements?	• The CAB has a temperature-controlled and humidity-monitored storage facility with a clear storage procedure outlined to control the temperature within the facility at 16 to 20 degrees to prolong and prevent accelerated shelf life(deterioration). Moreover, the preparation procedure outlines that reference material that enters the storage facility must have been clearly labelled giving reference to the requirements for reference material labels outlined in clause 7.14 of ISO 17034 for RM documents and labels, also addressed in Material handling and storage procedure		
	• Has the RMP defined procedures for transport to the customer?	• The CAB has defined a procedure for distribution and customer support, which outlines packaging specifications and certificates that accompany the materials distributed • See attached 7.15 Distribution procedure		
7.4.6	Has the RMP taken measures to ensure the integrity of the RM, until a seal is broken, or on initial usage?	• Material storage and handling to ensure integrity are addressed in the Reference Material Certificate, which is issued upon purchase of certified reference material via a customer-supplied email and accompanies the distributed reference material for easy access to pre-use, storage, and usage instructions outlined in the certificate. • See attached certificate for one of the reference materials		
7.5	**Material Processing**			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.5.1	• Has the RMP established procedures to ensure adequate processing of RM's? • Do these procedures address at least:-	• Procedure for the processing of reference material forms part of the sample preparation of coal sample procedure which highlights processes to be followed, sample handling and equipment pre use verifications. • See Procedure attached. This attachment addresses points a-i		
a)	Qualitative analysis to ensure verification of material type or identity?			
b)	Synthesis, purification, incubation and or transformation?			
c)	Homogenization?			
d)	Proper handling?			
e)	Measurements for control of the material processing?			
f)	Pre-treatment, cleaning or sterilization of equipment and sample containers?			
g)	Stabilization of the material?			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
h)	Packaging of the material?			
i)	Safety precautions?			
7.5.2	Is the equipment used in the processing operated in accordance with documented procedures? NOTE: Manufacturer's instructions are considered as documented procedures	• Personnel operating any equipment within the laboratory were trained, deemed competent and authorized to use laboratory equipment for processing and analysis of coal materials against established procedures. The procedures are derived from internationally recognized standards(ISO 1928, ISO 13909, ISO 562) and manufactures' equipment manuals. This is to ensure correct usage of equipment and ensure adherence to procedures and manufactures' manuals • See attached training records of one of laboratory personnel		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.6	• Are all the requirements of ISO/IEC 17025 met with respect to calibration & testing?	• The CAB addresses the requirements of testing and calibration outlined in ISO 17025 such that equipment(where applicable) employed within the organization which affect the quality and validity of results are calibrated by a SANAS accredited facility to ensure traceability to internationally recognized standards & SI units. Inhouse calibration is performed using calibration standards that are certified for compliance(benzoic acid pills for CV calibration). Testing of coal material is done following procedures that are derived from internationally recognized standards(ISO 1928, SANS 5925, ISO 589 etc). This also addresses traceability. The Lab uses certified reference materials of known value at outlined intervals to monitor any equipment drifts and if the equipment still produces acceptable, repeatable and reproducible results. • The lab enforces equipment verification at adequate intervals for equipment that does not require calibration to ensure that the equipment Is still fit for purpose prior to use. • See attached equipment register for calibration intervals, calibration certificates and equipment status. • Also attached, Calorific value Determination Procedure		
	• Are the measurements and calibrations consistent with the accuracy of the RM and any relevant standard specifications for the measurements?	• All measuring equipment are calibrated by a SANAS accredited facility and scrutiny of the certificates of calibration is done to ensure that equipment calibration is accurately done to cover our range of work. This assists in giving the CAB confidents when using such equipment to process reference materials. • See attached equipment register which includes calibration certificates of all measuring equipment		
7.7	**Measuring equipment:**			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	Is all the measuring equipment used in the RM production in compliance with the relevant clauses of ISO/IEC 17025?	• Clause 7.7 entails that all measuring equipment employed within the CAB have to be calibrated by an SANAS accredited facility and that the calibration must be traceable to internationals standards and SI units. All the appropriate equipment within the CAB follows the requirement outlined. The lab performs equipment verification, balance verifications prior to use to ensure that equipment is still fit for its intended purpose. • Certificates of calibration attached in the equipment register, also attached is the template employed for equipment verification and efficiencies		
7.8	**Data integrity and evaluation:**			
7.8.1	Does the RMP ensure that all calculations and data transfers are subject to appropriate checks?	• The CAB has in place a procedure used to validate all information systems(LIMS, excel spreadsheets, calculations, data transfers) to safeguard data integrity prior to use and at outlined intervals. The CAB has a master document register that serves as backup in case a said document is corrupted at point of use. Moreover, the CAB protects its operating systems and documents by backing up the entire system quarterly in the case of malware/virus attacks which would result in data loss. To protect unauthorized changes to documents, the CAB has unchangeable PDF documents at points of use and personnel have the parts of the system need to carry out their activities and only authorized personnel have access to the full system. • See attached data integrity procedure • **This addresses points raised in 7.8.2 as well**		
7.8.2	Has the RMP ensured that:			
a)	• Software developed in-house is validated and found adequate for use?			
	• Off-the-self software that is adapted is validated a found adequate for use?			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
b)	• Are procedures established and implemented for protecting the integrity of data?			
	• Do these procedures address data entry, capture, storage, transmission and processing?			
c)	• Is the data processing equipment maintained to ensure proper functioning?			
	• Are the environmental and operating conditions satisfactory for the maintenance of data integrity?			
d)	• Procedures established and implemented for the maintenance of data security?			
	• Do these procedures address unauthorized access, including amendments to records?			
	Note: Validation could include evidence of a sound theoretical basis – by reference to appropriate literature, and certain assumptions which can be shown to apply to the data.			
7.8.3	Are all statistical procedures used appropriate for their application?	• All statistical processes and procedures are derived from internationally recognized standards(ISO 13528, ISO 17034 & ISO 17043). These are used as the primary guideline when conducting any statistical process within the laboratory.		
7.9	**Metrological traceability of certified values**			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.9.1	Has the metrological traceability for all CRM's been established in accordance with ISO/IEC 17025?	• For gross calorific value determination, the CAB establishes metrological traceability through the use of ISO 1928 as the reference method for deriving its SOP. The calibration of the bomb calorimeter is performed using certified benzoic acid reference material, the assigned value of which is traceable to the SI unit of energy (joule, J) through the international thermochemical scale. All associated measuring instruments—including the analytical balance (for weighing samples and standards) and temperature sensors (for calorimeter water jacket and bomb temperature measurement)—are calibrated by SANAS-accredited laboratories with traceability to the SI units of mass (kg) and temperature (K). The CAB further demonstrates measurement comparability through participation in coal proficiency testing schemes and the routine use of certified reference materials for calibration verification. This ensures that all reported calorific values are metrologically traceable to the SI and conform to ISO/IEC 17025 and ISO 17034 requirements • For proximate analyses (moisture, ash, volatile matter), where direct traceability to SI units is not technically feasible due to the operational nature of the measurand which are operationally defined tests heavily reliant on the test method conditions, the CAB establishes metrological traceability through the use of standardized ISO and SANS methods (ISO 1171, ISO 562, SANS 5925), calibrated measurement equipment traceable to SI units (mass and temperature), and verification using certified coal reference materials and participation in inter-laboratory comparison. Furthermore, the CAB has validated its methods using certified reference materials with certified values. This ensures that all assigned values are traceable to appropriate reference systems in accordance with ISO/IEC 17025 and ISO 17034." • This is thoroughly documented in **6.5 Metrological traceability memorandum** • See attached equipment register with calibration certificates of equipment, method validation for one of the proximate as evidence of validation. **This addresses the entire 7.9 clause.**		

© SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	Is there evidence of metrological traceability of the certified value to a stated reference?			
7.9.2	Is the stated reference, by definition to a measurement unit (by practical realization), or a measurement procedure or a measurement standard?	• The CAB, through the development of its Certified Reference Material & Certificates, it references the standards with which its methods are derived from. This is evident in all of its CRM certificates, with one of it attached. The statement of conformity also has this covered. Each and every procedure derived from an international standard references the said standard. Calibration of equipment(where applicable) are done by an accredited facility and ones that are done inhouse are done using standards that are traceable to recognized SI units. • Find attached one of the procedures (CV), Statement of Conformity and CRM 002		
7.9.3	Where traceability to the SI is possible, is the RMP able to demonstrate this?	• Instruments involved in the characterization process are calibrated by a SANAS accredited facility with certificates of calibration demonstrating traceability to SI units, thereby keeping an unbroken chain of calibration. • See attached equipment register which has calibration certificates.		
7.9.4	Where traceability to the SI is not possible, is the RMP able to demonstrate traceability to an appropriate reference? These references could be i) CRM's from a competent producer ii) reference measurement procedures iii) specified methods or consensus standards (ISO/IEC 17025:2017)	• For proximate parameters (moisture, ash, volatile) traceability is established to the ISO reference method (ISO 1171 / ISO 562 / ISO 13909, SANS 5925) and verified using certified coal RMs during the validation of such methods. and PT participation is also another factor for maintaining the method; justification for non-SI traceability is recorded. • Find attached validations, ASH, IM, VOL.		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.9.5	Where traceability to a higher order reference system is required, is the RMP able to demonstrate that the measurements are calibrated with standards with metrological traceable values?	• where higher-order references are required we use SANAS -traceable standards and retain certificates demonstrating the calibration chain to the higher-order references. E.g equipment register with calibration certificates of calibration.		
7.9.6	Is the RMP able to provide evidence that secondary parameters having a significant influence on the certified value or its uncertainty are metrologically traceable?	• Secondary parameters (e.g., oven temperature, balance mass, furnace thermocouple) are listed in the equipment register and are covered by calibration certificates; their contributions are included in the uncertainty budget.		
7.10	**Assessment of homogeneity:**			
7.10.1	Has the RMP conducted the homogeneity testing of the candidate RM in its final packaging? Note: Assessment of homogeneity could include prior evidence, but experimental guide study would then be necessary. The homogeneity tests require a representative number of samples chosen at random, or systematic starting from a random start point.	• Given the nature of how the CAB processes its reference materials as stipulated primarily its preparation procedure, with the RM derived from a PT sample, homogeneity assessments are conducted prior to packaging of reference materials to ensure that each reference material is packaged with the confidence that the reference material is homogenized across all the packaged pulps. • Refer to Preparation Procedure attached & PT report which includes proof of homogeneity study under homogeneity assessment.		
7.10.2	Is the RMP able to provide evidence that when multiple batches are produced there is equivalence between the batches, or has the homogeneity of each batch been determined independently?	• As stipulated in the preparation procedure, multiple batches will only be made available after a sample goes for PT and after statistical studies conclude that the assigned values are reproducible to an already established CRM, then it can form a batch of that CRM which would mean that its homogeneity is determined independently. • See attached procedure under the Rationale for this explanation.		
7.10.3	• Have validated measurement procedures used to the homogeneity testing?	• The homogeneity studies form part of the value assignment statistics included in PT statistics. • See attached spreadsheet as proof of work, under the sheet named 'homogeneity'		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	• Is the selectivity and precision of the procedures fit for purpose?	• The decision rules on whether the homogeneity of a sample is sufficient are follows guidelines stipulated from ISO 13528(annexture B homogeneity and stability of proficiency test items) which render the procedures the CAB follows fit for purpose. This forms part of the derived Production Plan that covers parts that touch on CRM production.		
7.10.4	• Has the homogeneity for each property of interest been determined independently, or	• The homogeneity of a sample is done using only ash values, as ash has been studied and proven to be the most stable parameter to rely on; hence, the CAB only uses ash to determine the homogeneity of a sample.		
	• Can it be shown, scientifically or otherwise that measurement of one property is closely associated with other properties in the group?	• The CAB has demonstrated through homogeneity and ongoing stability studies that measurement of ash content, being the most stable and representative parameter of the coal matrix, adequately reflects the variability of other associated properties (such as volatile matter and calorific value). This scientific association is based on established correlations observed in the material and literature evidence showing ash content as a key indicator of coal composition. The CAB maintains a stability monitoring spreadsheet, and data will be evaluated in nine months to confirm that no significant changes occur in the ash or related parameters. This will confirm that the selected property (ash) is suitable as a representative measurand for homogeneity assessment, in accordance with ISO 17034:2016 Clause 7.10.4		
7.10.5	• Has the RMP for all certified values, quantified the homogeneity as an uncertainty contribution, or shown as a negligible contribution to the uncertainty?	• The CAB has quantified homogeneity & stability of a sample as an uncertainty contributor, and it includes it in the evaluation of uncertainties when assigning certified values for CRM's • See attached spreadsheet as evidence.		
7.11	**Assessment and monitoring of stability:**			
7.11.1	Does the RMP have:			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
a)	Access to the stability of all relevant properties of a RM under proposed storage conditions?	• The CAB is relatively new and is conducting its study through the stability monitoring of CRM's under the defined storage conditions(temperature range of 16 to 20 degrees Celsius) • Proof of study is attached.		
b)	Access to the stability of all relevant properties of a RM under proposed transport conditions?	• Stability due to Transportation effects is considered negligible as coal is not a volatile substance when exposed. But precautions are outline to safeguard the quality of materials when in transit.		
c)	Advice on storage and use necessary to maintain stability by the user?	• Upon receiving the certificate of the material, customers are advised to store the reference material in rooms that are under 23 degrees Celsius and upon opening, must use the material within 12 months. • See attached certificates as evidence		
d)	• Has a scheme been selected that allows for the monitoring the long-term stability?	• The stability monitoring of CRM is done on a monthly basis on CRMs produced and are covered in stability monitoring & procedure 7.10		
	• Does this scheme allow for prompt detection of changes to the stability?	• Procedure 7.10 states that if the criteria is not met, it triggers an investigation and possible re-evaluation of the CRM.		
	• Does the scheme take into account the possible rate of change?	• In the event that stability assessment fails, the CAB repeats the bottles for stability test. And if it fails, this would call for re-evaluation. Covered in 7.10		
e)	• Where the stability cannot be ensured has the RMP made allowance for changes in the uncertainty?	• During re-evaluation, the CAB also take into account the re-establishment of uncertainties as the establishment of such are derived from the homogeneity study stability study which will all be re-evaluated and new uncertainties established thereafter • See attached procedure 7.10.		
	• Or, where changes to the certified value can be predicted, a means of correcting the certified value and its uncertainty?	• In the event that a need arises for the CAB to re-evaluate its assigned values for an already produced CRM, the CAB may choose to re-evaluate the affected CRM by redistributing the said lot to its participants as a PT, conduct a new statistical analysis to assign new values and follow established prompts to communicate the amendments to its customers. • Find attached procedure 7.10 which outlines this.		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
f)	• Where repeated usage of a RM is permitted, has the RMP assessed the possible effects on the stability, and taken appropriate action?	• The most adversely affected CRM is that which is used continuously prone to environmental exposure. For this, the assessment of stability was specifically established in such a way that the CAB uses the same initially assessed bottles chosen at random in the initial stage of value characterization and assignment. This allows for dual assessment of stability in that the effects of multiple usage and environmental usage of the CRM is also evaluated.		
7.11.2	• Has the RMP conducted experimental assessment of the stability?	• The CAB is relatively new so it is still conducting its experimental assessment of the stabilities of CRM's produced and their assigned shelf life.		
	• Or does the RMP have evidence of stability from prior evaluation of similar batches?			
7.11.3	Where multiple batches of RM are produced, does the RMP evidence that sufficient stability testing between batches has been done, to provide confidence in the stability of all batches?	• The CAB has experimented on production of multiple batches and since decided that for a sample to be considered a batch to an already produced CRM it needs to have an independent PT statistical analysis of which its assigned values are reproducible to an already established CRM. This is outlined in AS001IW procedure under rationale.		
7.12	**Characterization:**			
7.12.1	Has the RMP assigned property values only after characterization has been conducted?	• Assignment of property values is only concluded after statistical analysis of data has been completed. How the CAB does this is by taking consensus values from a concluded PT round and conducts validation and uncertainties of parameters and producing a certificate thereafter.		
7.12.2	• Has the RMP defined whether the characterization is quantitative or qualitative property?	• Each property is defined operationally in the characterization and assignment of property values and their uncertainties (e.g., "volatile matter — ISO 562, % dry basis") and the nature (quantitative) is recorded		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	• And, if quantitative whether the measurand is defined operationally, or defined independently?	It is defined operationally by international coal ISO standards as stated in section 7 of procedure 7.13		
7.12.3	Has the RMP chosen a characterization strategy appropriate to its intended use? Note: characterization could follow one of these approaches a) single reference measurement procedure in a single lab, b) using two or more methods, in two or more laboratories, c) characterized in a network of competent laboratories, d) transfer from a closely matched RM using a single measurement procedure, or e) characterization based on mass or volume of ingredients used in the preparation.	• The CAB has chosen to follow a single reference measurement procedure and characterization in a network of competent laboratories as it uses its PT scheme as basis for its value assignment. Covered in 8.2. Experimental design requirements of procedure 7.13		
7.12.4	• Has the RMP specified the characterization study, such that the properties of interest have appropriate traceability and sufficient reliability?	• Details of azuresky approved plan is covered and responsible personnel are included; participating labs are competent (ISO/IEC 17025 accreditation / PT history) and documented. • These are covered in procedure 7.13 and example of a production plan also attached.		
a)	• Has the RMP documented a measurement plan that clearly describes the tasks to be performed, and is this plan communicated to responsible personnel?	• Covered in the production plan report.		
b)	• Where CRM's are certified, has the RMP demonstrated the competence of each laboratory by using data not obtained through the material to be characterized?	• Participating labs are governed by ISO 17025. Where a lab is not; it is the case where it is a competent onsite laboratory governed by a mother lab(Main Lab) accredited to ISO 17025		
7.12.5	• When evaluating the characterization data has the RPM performed a technical evaluation of the data and documents involved?	• Characterization data in its entirety is reviewed by the operations manager that is not involved in any way possible during the production of a CRM, who also approves the certificates of CRMs. • Find attached, CRM certificate for CRM 008, PT report • Also addresses the point below.		
	• Has adherence to the measurement plan been confirmed?			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	• If there have been deviations from the measurement plan, has the RMP decided if the deviation necessitates exclusion of the data from the characterization?	• Deviations to plans are documented and reviewed, and extent of the deviation will be the deciding factor to approve or reject the said characterization. Find attached, deviation report template.		
7.13	**Assignment of property values**			
7.13.1	Does the RMP have documented procedures for the assignment of property values?	• The assignment of property values follows the documented SOP 7.13 which outlines experimental design, statistics and uncertainty approach. • **The procedure attached addresses clause 7.13.2 a-e in its entirety**		
7.13.2	Do these procedures address:			
a)	• Details of the experimental designs?			
	• Details of the statistical techniques used?			
b)	• Policies for the treatment of outliers?			
	• Policies for the treatment and investigation of anomalous results?			
c)	• Weighing techniques for results from different laboratories, or procedures?			

©SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
d)	• The approach used to assign uncertainties to property values?			
e)	• Other factors that could influence the assignment of the property values?			
7.13.3	Has the RMP taken in account information on test methods, equipment, uncertainties reported, and evidence of past performance?	• Method validation was initially conducted on the methods selected and compatibility of instruments and their uncertainties and historic PT performance were considered during characterization and value assignment.		
7.13.4	• Have outliers been excluded without investigation of the possible reasons?	• Outliers are removed during the statistical evaluation under PT statistical analysis. To assign robust averages on CRM values and their standard deviations, the CAB employs Algorithm A of ISO 13528. Addresses the point below.		
	• Have robust techniques been used where appropriate?			
7.13.5	Has the RMP for all 'certified' values considered all uncertainty contributors to be included in the assigned uncertainty?	• The CAB addresses this clause with the uncertainty budget attached.		
7.13.6	For all 'certified' values has the RMP considered the following minimum contributors:-	• The procedure 7.13 outlines the contributors listed below and effects of some of the listed contributors are currently studied in the stability monitoring spreadsheet		
a)	• Characterization, including differences between multiple procedures?			

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
b)	• Between and within unit inhomogeneity?			
c)	• Changes in property values during storage?			
d)	• Changes in property values during transport?			
e)	NOTE: Where values other that certified values are assigned, a statement of uncertainty could be appropriate and improve the use of the material.			
7.14	**RM documents and labels:**			
7.14.1	• Has the RMP made available a certificate for all CRM's?	• All CRMs produced have a certificate which after purchase are made available to customers.		
	• Has a product information sheet been made available for all other RM?	• Product information is addressed within the CRM certificate. • Find attached CRM 002		
7.14.2	Do the certificates and product information sheets include at least the following?	• Find attached CRM 002 as evidence of compliance to the below list.		
a)	• Title of the document?			
b)	• Unique identifier of the RM?			

©SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
c)	• The name of the RM?			
d)	• Name and contact details of the RMP?			
e)	• intended use			
f)	• Minimum sample size (where applicable)?			
g)	• Period of validity?			
h)	• Storage information?			
i)	• Instructions on handling?			
j)	• Page number / total number of pages?			
k)	• Document version (number)?			
l)	• Information on commutability?			

©SANAS

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
	VIM Definition – commutability of a reference material: property of a reference material, demonstrated by the closeness of agreement between the relation among the measured results for a stated quantity in this material, obtained according to two given measurement procedures, and the relation obtained among the measurement results for other specified materials. (Particularly of interest for clinical RM)			
7.13.4	In addition to a) to f) above do the certificates for CRM's also include:			
a)	• A description of the CRM?	• CRM 008 is attached as proof of compliance to the list below.		
	• Property of interest, property value and uncertainty?			
	• Measurement procedure for operational defined measurands?			
	• Metrological traceability of the certified values?			
	• Name and function of the person responsible for the approval?			
7.14.4	• Is the RM label securely attached to individual RM container?	• The CAB uses durable label stickers from a trusted supplier to securely label its RM containers.		
	• Is the design such that it will remain legible for the defined handling and storage conditions?	• The design is made to remain attached securely to the RM bottle, so it lasts the defined handling and storage conditions		
	• Does the label identify the material, the RM, the batch and any other pertinent information, such as a unique sample number?	• All relevant information pertaining to the CRM produced is addressed visibly on the CRM label		

CLAUSE	ISO/IEC 17034:2016 REQUIREMENTS	CAB's COMMENTS	C/ NC	ASSESSOR's COMMENTS
7.14.5	• In cases where size does not permit all required information on the label, is it included elsewhere, and linked by a unique identifier?	• Any other information relevant to the CRM Is addressed on CRM certificate which is issued on purchase. • Find attached CRM label checklist the CAB follows to ensure compliance to the above list		

Additional / General Comments *This space may also be used to expand on comments in specific sections*

Team Leader's Signature:		**Date:**	

©SANAS